TRACE — Pipeline Architecture



FEATURE EXTRACTION

OCR / Text	UI / YOLO	CNN Visual	Interaction	Temporal
50	30	1,280	30	1
EasyOCR	YOLOv8 detection	EfficientNet-B0	click · key · drag	flag / context

Combined feature vector: **1,391 dimensions**

TIER 1 — Classical ML

SVM
KNN
Random Forest
Decision Tree
XGBoost · LightGBM

scikit-learn

TIER 2 — Deep Learning

LSTM
CNN1D
MLP · Transformer

PyTorch

TIER 3 — Ensemble

Stacking
(SVM + RF + MLP → meta)

Voting (hard + soft)

Best F1 = 0.89

scikit-learn

Activity Label — 17 Classes

coding · debugging · terminal · web_browsing · git_ops · docker
media_editing · diagram_design · image_editing · file_management · …

Knowledge Generation (Rule-Based)

SOP · Runbook · Timeline · RAG Search